

Indian Statistical Institute

End Semester Examination (Part I)

Optimization and Numerical Methods
20 May 2026, 2:00 pm – 3:30 pm

Bachelor of Statistical Data Science
Maximum marks in Part I: 27

INSTRUCTIONS

- The exam is divided into two parts: Part I and Part II. Each part carries a maximum of 27 marks. However, note that the total marks that can be scored for this exam, combining both the parts, are capped at 50.
- Part I contains three questions. Answer ALL the questions. The marks assigned to each question are indicated alongside the question.
- Use of scientific calculator is permitted. No other electronic devices are allowed in the examination hall.

QUESTIONS

1. (a) Define a convex optimization problem. Define the feasible set S of a convex optimization problem and show that S is a convex set. [3]
(b) State the first-order optimality condition for a feasible point x^* to be an optimal solution of a differentiable convex optimization problem. [2]
(c) Consider the following optimization problem:

$$\begin{aligned} &\text{minimize} && f(x, y) = x^2 + y^2 - 4x - 2y \\ &\text{subject to} && x + y \leq 1 \\ &&& x, y \geq 0 \end{aligned}$$

Show that the above problem is a convex optimization problem. Further, verify using the first-order optimality condition that $x^* = (1, 0)^T$ is an optimal solution. [4]

2. (a) The Lumina Lighting Company produces two types of high-end LED fixtures: Aura (x) and Radiant (y). The company wishes to determine the daily production quantities that maximize profit. Each Aura yields a profit of Rs. 100, while each Radiant yields a profit of Rs. 120. The production process is subject to the following constraints:
 - The Assembly Department provides at most 54 labour-hours per day. Each Aura requires 2 labour-hours, and each Radiant requires 1 labour-hour.
 - The Polishing Department provides at most 48 labour-hours per day. Each Aura requires 1 labour-hour, and each Radiant requires 2 labour-hours.
 - Due to quality-control bottlenecks, at most 30 total units can be produced per day.

Formulate the above problem as a Linear Programming Problem and solve it graphically. [4]

(b) Consider the following optimization problem:

$$\begin{aligned} &\text{minimize} && f(x, y) = e^x + e^{2y} \\ &\text{subject to} && x + 2y \geq 8 \end{aligned}$$

Show that this is a convex optimization problem and solve it using KKT conditions. [5]

3. (a) Let $(x_0, y_0), \dots, (x_k, y_k)$ be data points with distinct x_i and let $P_k(t)$ be the interpolating polynomial of degree at most k passing through all these points. Suppose (x, y) is a new data point with $x \neq x_i$ for every $i = 0, \dots, k$ and let $P_{k+1}(t)$ be the new interpolating polynomial of degree upto $k + 1$ passing through all the points.

(i) Show that for some constant C ,

$$P_{k+1}(t) = P_k(t) + C \prod_{i=0}^k (t - x_i).$$

[2]

(ii) Using $P_{k+1}(x) = y$, compute the value of C in terms of $P_k(x), y$ and x_i . Hence show that: [2]

$$P_{k+1}(t) = P_k(t) + (y - P_k(x)) \prod_{i=0}^k \left(\frac{t - x_i}{x - x_i} \right)$$

- (b) (i) Compute the quadratic interpolating polynomial passing through $(-1, 1), (0, 1)$ and $(1, 3)$ using Lagrange interpolation. [3]
(ii) Suppose a new data point $(2, 13)$ is added. Using the expression obtained in (a) and the quadratic polynomial obtained previously, compute the cubic polynomial passing through all the four points. [2]